

MIR SAHIN ALI

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SUMMARY

Final-year B.Tech Computer Science Engineering student with hands-on experience in software development, component design, REST API integration, unit testing, debugging, SQL databases, and technical documentation. Proficient in Java, Python, JavaScript, SQL, Git, GitHub, Docker, and software engineering best practices. Familiar with Agile SDLC, object-oriented design, design patterns, system integration, testing frameworks, and AI-assisted development. Focused on building reliable, maintainable, and high-quality software solutions that meet functional and technical requirements.

SKILLS

Programming Languages: Java, Python, C, JavaScript, SQL

Software Development: Object-Oriented Programming, Component Design, Modular Development, Design Patterns, Debugging, Error Handling

Testing & Quality Engineering: Unit Testing, Integration Testing, API Testing, JUnit, PyTest, Test Case Design, Functional Testing, Software Quality Assurance

Web & System Integration: REST APIs, JSON, HTTP, React Native, Expo, HTML, CSS, JavaScript, API Integration, Data Interfaces

Database: PostgreSQL, SQL, Relational Databases

SDLC & Methodologies: Software Development Life Cycle (SDLC), Agile, Scrum, Waterfall, Requirements Analysis, Technical Design, Implementation, Testing, Documentation, Maintenance

Developer Tools: Git, GitHub, GitHub Actions, Docker, VS Code, Linux, Postman

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Software Engineering

PROJECTS

Modular Task Management & Workflow Automation System

Tech Stack: Java, Spring Boot, REST APIs, MySQL, JUnit, Mockito, Git, Docker

- Designed and developed a modular task-management information system using layered architecture and reusable software components to support task creation, assignment, status tracking, prioritization, and user workflows.
- Created functional and technical component designs covering API contracts, database models, service responsibilities, validation rules, and interactions between application modules.
- Developed RESTful APIs using Spring Boot and integrated controller, service, repository, and database components into a cohesive information system based on defined functional and technical requirements.
- Applied object-oriented programming principles, SOLID practices, dependency injection, DTO patterns, exception handling, and reusable service components to improve maintainability and separation of concerns.
- Implemented unit tests using JUnit and Mockito for service and controller components, covering business logic, validation rules, exception scenarios, and API behavior.
- Performed debugging, integration testing, and API testing with Postman, and documented endpoints, component behavior, test cases, setup procedures, and implementation decisions for future maintenance.
- Followed an Agile-oriented SDLC workflow using Git and GitHub for requirements analysis, feature development, version control, code review, testing, documentation, and iterative delivery.
- Containerized the application using Docker and maintained structured Git branches and GitHub Actions workflows to automate build and test verification.

Service Integration & Order Processing Platform

Tech Stack: Python, FastAPI, REST APIs, PostgreSQL, PyTest, Docker, GitHub Actions

- Developed a component-based order-processing platform that integrates customer, product, inventory, and order services through REST APIs and structured data interfaces.
- Designed functional specifications, technical architecture, database schemas, API contracts, request-response models, and component interaction flows before implementation.
- Built independent software components for order validation, inventory verification, pricing, order status management, and transaction processing, then assembled them into an integrated information system.
- Implemented REST APIs using FastAPI with request validation, exception handling, structured JSON responses, status codes, and reusable service-layer logic.
- Applied software design principles and patterns including separation of concerns, repository pattern, service-layer abstraction, dependency management, and modular component design.
- Created unit and integration tests using PyTest to validate API endpoints, business rules, database operations, component interfaces, error conditions, and end-to-end order workflows.
- Prepared detailed technical documentation covering system architecture, component responsibilities, API specifications, database design, test cases, deployment instructions, and maintenance procedures.

- Used Git, GitHub, Docker, and GitHub Actions throughout the SDLC for version control, collaborative development, automated testing, integration, and reproducible application deployment.

INTERNSHIPS

DokLink Services Pvt. Ltd. – Frontend Development Intern

June 2025 – July 2025

Tech Stack: React Native, Expo, REST APIs, JavaScript

- Contributed to the development of a Hospital Bed Management System providing real-time hospital bed availability, hospital search, booking functionality, and patient-facing workflows.
- Integrated REST APIs with React Native interfaces to retrieve, process, and display dynamic backend data across multiple application screens.
- Implemented form validation, state management, loading indicators, error handling, authentication workflows, and navigation.
- Developed responsive and reusable UI components while maintaining consistency with application design requirements.

Employees' Provident Fund Organisation (EPFO) – Admin & Data Support Intern

May 2025

Tech Stack: Data Management, Documentation, Internal Software

- Supported data digitization, documentation, compliance tracking, and structured administrative information-management activities.
- Worked with internal software systems and digital records while maintaining data accuracy and documentation quality.

EDUCATION

Amity University Kolkata

B.Tech in Computer Science and Engineering; CGPA: 8.01/10

Kolkata, West Bengal

2023 – 2027

Chowberia Dinabandhu Vidyalaya

Class XII: 80.80% | Class X: 74.71%

Kolkata, West Bengal

2021 | 2019

CERTIFICATIONS

- **Java Programming** – Udemy
- **Foundations of Cybersecurity** – Coursera
- **Manage Security Risks** – Coursera
- **Geospatial Technology for Climate Smart Agriculture** – ISRO

ACHIEVEMENTS

- **GATE 2026 Qualified** – Graduate Aptitude Test in Engineering.
- **Badminton Player** – Active badminton player with participation in competitive and recreational matches.